GLANSIS ORGANISM RISK ASSESSMENT

This work may be cited as: Shelly, C. 2023. Risk Assessment for *Limnoperna fortunei*. NOAA-GLERL: GLANSIS.

SPECIES INFORMATION

Scientific Name: *Limnoperna fortunei* (Dunker, 1857)

Common Name: Golden mussel

SECTION A: INTRODUCTION POTENTIAL RESULTS

Dispersal: Unlikely
Hitchhiking/Fouling: Unlikely
Unauthorized intentional release: Unlikely
Stocking/Planting/Escape from recreational culture: Unlikely
Escape from commercial culture: Unlikely
Shipping: Moderate

***Limnoperna fortunei* has a moderate probability of introduction to the Great Lakes (Confidence level: High).**

Potential pathway(s) of introduction: Transoceanic Shipping

Comments:

Limnoperna fortunei has a high tolerance to fluctuating salinities; no significant mortality was observed in mussels exposed to a salinity cycle with abrupt salinity changes ranging 1–23% (mean 2.68%) over a month (Sylvester et al. 2013). This will affect the probability of uptake (many ports located in estuarine environments with fluctuating salinity) and survival after ballast exchange (Sylvester et al. 2013). Though tolerance of larvae has not been specifically examined, larval survival is likely due to presence of colonies in saline regions (unlikely formed by drifting adults) (Sylvester et al. 2013). Ship traffic to the Great Lakes includes areas already invaded by *L. fortunei*, such as South America and Asia (Boltovskoy 2015, EPA 2015). While ship traffic from these regions is not frequent, the potential for introduction exists.

POTENTIAL INTRODUCTION VIA DISPERSAL

A1) Does this species occur near waters (natural or artificial) connected to the Great Lakes basin* (e.g., streams, ponds, canals, or wetlands)? (*Great Lakes basin = below the ordinary high water mark, including connecting channels, wetlands, and waters ordinarily attached to the Lakes)

Yes, this species occurs near waters connected to the Great Lakes basin and is mobile or able to be transported by wind or water.	100
No, this species does not occur near waters connected to the Great Lakes basin and/or is not mobile or able to be transported by wind or water.	0 √
Unknown	U

- Limnoperna fortunei *does not occur near waters connected to the Great Lakes basin.*

A2) What is the proximity of this species to the Great Lakes basin?

This species occurs in waters within 20 kilometers of the Great Lakes basin, and no barrier (e.g., electric barrier, dam) to dispersal is present.	Score x 1
This species occurs in waters within 20 kilometers of the Great Lakes basin, but dispersal to the basin is blocked; or, this species occurs in waters within 100 kilometers of the Great Lakes basin, and no barrier to dispersal is present.	Score x 0.75
This species occurs in waters within 100 kilometers of the Great Lakes basin, but dispersal to the basin is blocked.	Score x 0.5
This species occurs in waters >100 kilometers from the Great Lakes basin.	Score x 0.25
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA HITCHHIKING/FOULING

A3) Is this species likely to attach to or be otherwise transported by, or along with, recreational gear, boats, trailers, fauna (e.g., waterfowl, fish, insects), flora (e.g., aquatic plants), or other objects (e.g., packing materials), including as parasites or pathogens, entering the Great Lakes basin?

Yes, this species is known to be able to adhere to certain surfaces or to be transported by other organisms entering the Great Lakes basin.	100
No, this species is not known to be able to adhere to certain surfaces or to be transported by other organisms entering the Great Lakes basin.	0 √
Unknown	U

- Limnoperna fortunei *may have been introduced to Japan as contaminant of shipments of live golden mussels from China (Crosier and Molloy 2004, Ghabooli et al. 2013, Magara et al. 2001), but golden mussels are not known to be shipped to the Great Lakes region.*
- *When structures fouled by* L. fortunei *are transported over land, this species can survive air exposure for 2 to 10 days depending on ambient temperatures (De Andrade et al. 2021). Given their current distribution, individuals are not capable of surviving an overland trip to the Great Lakes.*

A4) What is the proximity of this species to the Great Lakes basin?

This species occurs in waters within 20 km of the Great Lakes basin.	Score x 1
This species occurs in waters within 100 km of the Great Lakes basin.	Score x 0.5
This species occurs in waters >100 km from the Great Lakes basin.	Score x 0.1
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA UNAUTHORIZED INTENTIONAL RELEASE

A5) Is this species sold at aquarium/pet/garden stores ("brick & mortar" or online), catalogs, biological supply companies, or live markets (e.g., purchased for human consumption, bait, ornamental, ethical, educational, or cultural reasons) and as a result may be released into the Great Lakes basin?

Yes, this species is available for purchase.	100
No, this species is rarely/never sold.	0 √
Unknown	U

- *At the time of this review, the golden mussel is not sold in the Great Lakes basin, nor is it readily available for purchase online.*

A6) How easily is this species obtained within the Great Lakes region (states/provinces)?

This species is widely popular, frequently sold, and/or easily obtained within the Great Lakes region.	Score x 1
This species is widely popular, and although trade, sale, and/or possession of this species is prohibited, it is frequently sold on the black market within the Great Lakes region.	Score x 0.5
This species is not very popular or is not easily obtained within the Great Lakes region.	Score x 0.1
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA STOCKING/PLANTING OR ESCAPE FROM RECREATIONAL CULTURE

A7) Is this species being stocked/planted to natural waters or outdoor water gardens around the Great Lakes region?

Yes, this species is being stocked/planted and/or has ornamental, cultural, medicinal, environmental (e.g., biocontrol, erosion control), scientific, or recreational value in the Great Lakes region.	100
No, this species cannot be stocked/planted or there is not enough interest to do so in the Great Lakes region.	0 √
Unknown	U

- *At the time of this review, the golden mussel is not stocked or planted in the Great Lakes basin.*

A8) What is the nature and proximity of this activity to the Great Lakes basin?

This activity is authorized and/or is occurring directly in the Great Lakes.	Score x 1
This activity is occurring in Great Lakes tributaries or connecting waters, or within 20 km of the Great Lakes basin.	Score x 0.75
This activity is likely to occur within 20 km of the Great Lakes basin because of its popularity/value and there are no widespread regulations against stocking/planting.	Score x 0.5
This activity is occurring in waters >20 km from the Great Lakes basin, or despite federal or state regulations in more than half the basin (> 5 states/provinces), this activity may occur within 20 km of the basin because of the species' popularity/value.	Score x 0.25
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA ESCAPE FROM COMMERCIAL CULTURE

A9) Is this species known to be commercially cultured in or transported through the Great Lakes region?

Yes, this species is being commercially cultured in or transported through the Great Lakes region.	100
No, this species is not commercially cultured in or transported through the Great Lakes region.	0 √
Unknown	U

- *At the time of this review, there is no commercial culture or transport of golden mussels in the Great Lakes region.*

A10) What is the nature and proximity of this activity to the Great Lakes basin?

This activity is unregulated or minimally regulated and is occurring directly in the Great Lakes.	Score x 1
This activity is unregulated or minimally regulated and is occurring in Great Lakes tributaries or connecting waters, or within 20 km of the Great Lakes basin.	Score x 0.75
This activity is strictly regulated but occurs directly in the Great Lakes, and/or this activity involves transport of live organisms on/across the Great Lakes.	Score x 0.5
This activity is strictly regulated but occurs in Great Lakes tributaries, connecting waters, or within 20 km of the Great Lakes basin, and/or this activity involves transport of live organisms within 20 km of the Great Lakes basin.	Score x 0.25
This activity occurs >20 km from the Great Lakes basin and typically does not involve transport of live organisms closer to the basin.	Score x 0.1
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA SHIPPING

A11) Is this species capable of surviving adverse environments (i.e. extreme temperatures, absence of light, low oxygen levels) and partial-to-complete ballast water exchange (BWE) (e.g., is euryhaline,

buries in sediment, produces resistant resting stages, has other attributes or behaviors facilitating survival under these conditions)?

Yes, this species is able to survive in ballast tank environments for weeks at a time and may be suspended in ballast water.	100
Yes, this species is able to survive in ballast tank environments for weeks at a time and is able to survive BWE by burial in ballast sediment.	80 √
Yes, this species is able to survive in ballast tank environments for weeks at a time and may be suspended in ballast water, but this species is not able to survive BWE.	60
No, but this species is capable of fouling transoceanic ship structures (e.g., hull, chains, chain locker) while in its active or resting stage.	40
No, this species is not able to survive adverse environments, does not foul transoceanic ship structures, or is unlikely to be taken up with ballast.	0
Unknown	U

- Limnoperna fortunei *has a high tolerance to fluctuating salinities; no significant mortality was observed in mussels exposed to a salinity cycle with abrupt salinity changes ranging 1–23 % (mean 2.68 %) over a month (Sylvester et al. 2013). Tolerance to this type of regime was unaffected by different temperatures within ambient ranges, and survival was 90% over 9 hours (Sylvester et al. 2013). In terms of the distance covered by a commercial vessel typically sailing at 13–24 knots (24–44 km/h), 9 h represents anywhere between 217 and 400 km, which in many areas exceeds the distance between the destination port and the site of mid-ocean exchange (Sylvester et al. 2013).*
- *This mid-ocean ballast exchange will affect the probability of* L. fortunei *uptake (many ports located in estuarine environments with fluctuating salinity) and survival after ballast exchange (Sylvester et al. 2013). Though tolerance of larvae has not been specifically examined, larval survival is likely due to presence of colonies in saline regions (unlikely formed by drifting adults) (Sylvester et al. 2013).*
- L. fortunei *may also enter the US from South American ships visiting ports along the Gulf of Mexico. Due to their salinity tolerance, veligers are expected to survive the two week voyage. Ballast water regulations may reduce this risk but the salinity tolerance of* L. fortunei *may enable its larvae to remain in ballast water after exchange (Ricciardi 1998).*

A12) Does this species occur in waters from which shipping traffic to the Great Lakes originates?

Yes, and this species has been observed in the ballast of or fouling ships entering the Great Lakes.	Score x 1
Yes, and this species has been observed in ports that have direct trade connections with the Great Lakes (e.g., Baltic Sea).	Score x 0. 5 √
Yes, but this species has neither been observed in ballast/fouling ships entering the Great Lakes nor in ports in direct trade with the Great Lakes.	Score x 0.1
No, this species does not occur in waters from which shipping traffic to the Great Lakes originates.	Score x 0
Unknown	U

- *Ship traffic to the Great Lakes includes areas already invaded by* Limnoperna fortunei*, such as South America and Asia (EPA 2015).*
- L. fortunei *has been observed in ballast waters and successfully invaded South America through this means of introduction (Boltovskoy 2015).*
- L. fortunei *has been observed in ballast water and is distributed in proximity of ports with a direct trade connection with the Great Lakes (i.e. Brazil) (Boltovskoy 2015, EPA 2015).*

Potential Vector Ranking and Points

Vector	Raw Points Scored	Proximity Multiplier	Total Points Scored	Probability of Introduction
Dispersal: Natural dispersal through waterbody connections or wind	0	x	0	Unlikely
Hitchhiking/Fouling: Transport via recreational gear, boats, trailers, mobile fauna, stocked/planted organisms, packing materials, host organisms, etc.	0	x	0	Unlikely
Release: Unauthorized intentional release of organisms in trade (e.g., aquaria, water gardens, live food)	0	x	0	Unlikely
Stocking/Planting/Escape from recreational culture: Intentional authorized or unauthorized introduction to natural waters in the Great Lakes *OR* Accidental introduction to Great Lakes by escape from recreational culture (e.g., water gardens)	0	x	0	Unlikely
Escape from commercial culture: Accidental introduction to Great Lakes by escape from commercial culture (e.g., aquaculture)	0	x	0	Unlikely
Trans-oceanic shipping: Ballast (BOB) or no-ballast-on-board (NOBOB) water exchange/discharge, sediment discharge, hull fouling	80	x 0.5	40	Moderate
Total Unknowns (U) 0	Confidence Level		High	

Qualitative Statements for GLANSIS Fact Sheet:

Limnoperna fortunei **has a moderate probability of introduction to the Great Lakes (Confidence level:**
High).

Potential pathway(s) of introduction: Transoceanic Shipping

Scoring	
Points (per vector)	**Probability for Introduction**
80-100	High
40-79	Moderate
1-39	Low

0	Unlikely
# of Unknowns (overall)	**Confidence Level**
0	High
1-2	Moderate
3-5	Low
>5	Very low

SECTION B: RISK OF ESTABLISHMENT RESULTS

INVASIVE BIOLOGICAL/ECOLOGICAL ATTRIBUTES

B1) How would the physiological tolerance of this species (survival in varying temperature, salinity, oxygen, and nutrient levels) be described?

This species has broad physiological tolerance. It has been reported to survive in wide ranges of temperature (0°C-30°C), salinity (0-16 parts per thousand), oxygen (0-saturated), AND nutrient (oligotrophic-eutrophic) levels.	9
This species has somewhat broad physiological tolerance. It has been reported to survive in a wide range of temperature, salinity, oxygen, OR nutrient levels. Tolerance to other factors is narrower, unknown, or unreported.	6
This species has narrow physiological tolerance. It has been reported to survive in limited ranges of temperature, salinity, oxygen, and nutrient levels.	3
Unknown	U
	9

- Limnoperna fortunei can *survive (90%) up to a salinity shock of 2 ppt for periods of at least 10 days (Angonesi et al. 2008).*
- *The golden mussel has a high tolerance to fluctuating salinities; no significant mortality was observed in golden mussels exposed to a salinity cycle with abrupt salinity changes ranging 1–23 % (mean 2.68 %) over a month. Tolerance to this type of regime was unaffected by different temperatures within ambient ranges (Sylvester et al. 2013).*
- *Deaton et al. (1989) had 41% golden mussel survival in 800mOsm (seawater ~1000mOsm) water.*
- *Compared to* Dreissena polymorpha*,* Limnoperna fortunei *has higher resistance to anoxia, pollution (including eutrophication), pH, and high temperatures, longer reproduction periods and lower calcium requirements (3-4mg/L) (Karatayev et al. 2007). This broader tolerance indicates this species could have an even broader distribution in the Great Lakes than* D. polymorpha*.*

B2) How likely is it that any life stage of this species can overwinter in the Great Lakes (survive extremely low levels of oxygen, light, and temperature)?

Likely (This species is able to tolerate temperatures under 5°C and oxygen levels ≤0.5 mg/L)	9
Somewhat likely (This species is able to tolerate some of these conditions OR has adapted behaviorally to avoid them)	6
Somewhat unlikely (This species is able to tolerate conditions close to those specified, but it is not known as an overwintering species)	3
Unlikely	0
Unknown	U
	9

- Limnoperna fortunei *populations in Japan survive winter water temperatures of 5-6°C. Reproducing populations have been documented in the Paldang Reservoir, Korea which has water temperatures as low as 0°C (Karatayev et al. 2007).*

- L. fortunei *overwinters in South Korea, with water temperature as low as 0°C (Oliveira et al. 2011). In Japan, minimum temperature in a reservoir with golden mussels was 4.2°C (Nakano et al. 2011). Experimental research supports the 5°C threshold for prolonged exposure (Oliveira et al. 2011).*
- *Based on broad thermal tolerances, Ricciardi (1998) predicts* L. fortunei *colonization into the lower North American Great Lakes.*
- L. fortunei *can persist at low water temperatures (5°C) for several days and is capable of maintaining its energy reserves (glycogen) at this temperature (Andrade et al. 2018).*
- *Experiments to examine* L. fortunei *overwintering survival found that populations at the northern invasion front can survive 6 days, 41 days, and 108 days at <1°C, <2°C, and <5°C respectively. Overall survival was 27% at these temperatures. An accompanying species distribution model implies suitable habitats at higher latitudes than previously considered (Xia et al. 2021).*
- *An experiment examining* L. fortunei *anoxia tolerance reports a higher resistance to oxygen deprivation than the zebra mussel* Dreissena polymorpha. *Under laboratory conditions,* L. fortunei *was exposed to conditions with less than 0.16±0.02 mg/L dissolved oxygen and 100% sample mortality occurred after 10 to 30 days of exposure depending on mussel size and temperature (Perepelizin and Boltovskoy 2011).*

B3) If this species is a heterotroph, how would the flexibility of its diet be described?

This species is a dietary generalist with a broad, assorted, AND flexible diet.	9
This species is moderately a dietary generalist with a broad, assorted, OR flexible diet.	6
This species is a dietary specialist with a limited and inflexible diet.	3
This species is an autotroph.	0
Unknown	U
	7

- L. fortunei *consumes a variety of phytoplankton and zooplankton (Frau et al. 2012, Rojas Molina et al. 2012, Rojas Molina et al. 2010). Adults do well despite low food availability (Oliveira et al. 2011).*
- *Invasive golden mussel populations in Brazil prey on juvenile Daphniidae and Calocalanidae micro crustaceans and Vaginicolidae and Centropyxidae protozoa (Bertao et al. 2021).*
- *Through an isotope analysis, particulate organic matter was identified as the primary food source for* L. fortunei, *plankton and sediment organic matter were also major dietary components* (Zhang et al. 2017).

B4) How likely is this species to outcompete species in the Great Lakes for available resources?

Likely (This species is known to have superior competitive abilities and has a history of outcompeting other species, AND/OR available literature predicts it might outcompete native species in the Great Lakes)	9
Somewhat likely (This species is known to have superior competitive abilities, but there are few reported cases of this species outcompeting another and no predictions regarding species in the Great Lakes)	6
Somewhat unlikely (This species has average competitive abilities, and there are no reported cases of this species outcompeting another and no predictions regarding species in the Great Lakes)	3
Unlikely (This species is known as a poor competitor that thrives only in environments with low biodiversity, AND/OR available literature predicts it might be outcompeted by a species in the Great Lakes)	0
Unknown	U
	8

- Limnoperna fortunei *negatively affects burrowing invertebrates and unionids (Karatayev et al. 2010) in South America, and may do the same in the Great Lakes given if in high densities. Such competitive exclusion is also seen in a similar species,* D. polymorpha*.*
- L. fortunei *may have outcompeted/overgrown the native gastropods in South America,* Chilina fluminae *and* Biomphalaria straminea*, as they became rare and eventually disappeared from study sites (Spaccesi and Capitulo 2012).*
- L. fortunei *negatively impacts unionid species through competition for food. L. fortunei has been shown to physically foul living unionids through settlement on their shells. This impairs movement and restricts their ability to open their valves, thus depriving them of food and oxygen (Weigand 2019).*
- *The shell production by* Limnoperna fortunei *decreases flow velocity, increases shear strength and increases sedimentation. These ecosystem engineering effects lead to spatial interference competition with native bivalves, reed roots, etc. (Darrigan and Damborenea 2011).*
- *In mesocosm experiments,* Limnoperna fortunei *reduced the density of zooplankton through competition for trophic resources. Rotifer and cladoceran densities showed significant declines in mesocosms containing golden mussel (Rojas Molina et al. 2012).*

B5) How would the fecundity of this species be described relative to other species in the same taxonomic Class?

Very high	9
High	6
Moderate	3
Low	0
Unknown	U
	6

- *No data was found on natural* L. fortunei *fecundity, though very high rates of colonization suggest it is high (Karatayev et al. 2007).*
- *In the laboratory,* L. fortunei *induced to spawn produced approximately 11,000 eggs (n = 10, range 133-29,800 eggs) while* D. polymorpha *produces larger quantities of eggs per reproductive season, from 275,000-300,000 up to 1,000,000 eggs (Cataldo and Boltovskoy, unpublished data in Karatayev et al. 2007; Lvova 1977; Sprung 1991).*
- L. fortunei *spawns continually, while* D. polymorpha *is a batch spawner, which may contribute to the former's success (Boltovskoy et al. 2006).*
- *Analysis of a* L. fortunei *population in Brazil found a high annual growth rate (K = 1.22) and estimated that 62.920 juveniles/m2 will be recruited annually (Ayroza et al. 2021).*
- *The fecundity of native unionids is also high, for example the native mussel* Amblema plicata *has a reported fecundity of 75,000 to 500,000 eggs per reproductive season for adults 5 to 10 years old (Morales et al. 2007). However,* A. plicata *and other native unionids produce glochidia that require a suitable fish host to develop into adults.* L. fortunei *is capable of external fertilization and its larvae do not depend on a host for development (Boltovskoy et al. 2015).*

B6) How likely are this species' reproductive strategy and habits to aid establishment in new environments, particularly the Great Lakes (e.g., parthenogenesis/self-crossing, self fertility, vegetative fragmentation)?

Likely (The reproductive strategy or habits of this species are known to aid establishment in new environments, AND available literature predicts establishment in the Great Lakes based on these attributes)	9

Somewhat likely (The reproductive strategy or habits of this species are known to aid establishment in new environments, but there is no literature available regarding establishment in the Great Lakes based on these attributes)	6
Somewhat unlikely (The reproductive strategy or habits of this species could potentially aid establishment in new environments, but there is no literature available regarding establishment in the Great Lakes based on these attributes)	3
Unlikely (The reproductive strategy or habits of this species are not known to aid establishment in new environments)	0
Unknown	U
	6

- L. fortunei *spawns continually (Boltovskoy et al. 2006).*
- L. fortunei *reproductive strategies include fast growth, rapid maturity, recruitment to established colonies, and external fertilization that produces planktonic veligers (Boltovskoy et al. 2015). These strategies are similar to other invasive species that are highly successful in the Great Lakes (i.e.* Dreissena polymorpha *and* D. bugensis*).*

ENVIRONMENTAL COMPATIBILITY

B7) How similar are the climatic conditions (e.g., air temperature, precipitation, seasonality) in the native and introduced ranges of this species to those in the Great Lakes region?

Very similar (The climatic conditions are practically identical to those of the Great Lakes region)	9
Similar (Many of the climatic conditions are similar to those of the Great Lakes region)	6
Somewhat similar (Few of the climatic conditions are similar to those of the Great Lakes region)	3
Not similar	0
Unknown	U
	6

- *While the climate in South America is quite mild, South Korea's climate is similar to the Great Lakes. South Korea has a humid continental climate and a humid subtropical climate. Winters can be extremely cold with the minimum temperature dropping below −20 °C (−4 °F) in the inland region of the country: in Seoul, the average January temperature range is -7 to 1 °C (19 to 34 °F), and the average August temperature range is 22 to 30 °C (72 to 86 °F).*

B8) How similar are other abiotic factors that are relevant to the establishment success of this species (e.g., pollution, water temperature, salinity, pH, nutrient levels, currents) in the native and introduced ranges to those in the Great Lakes?

Very similar (These factors are practically identical to those of the Great Lakes region)	9
Similar (Many of these factors are similar to those of the Great Lakes region)	6
Somewhat similar (Few of these factors are similar to those of the Great Lakes region)	3
Not similar	0

Unknown	U
	7

- *Levels of these abiotic factors are very similar between the Great Lakes and the native/introduced ranges. That is, like the Great Lakes, the native/introduced ranges include extremes of pollution, water temperature, pH and nutrient levels. Salinity is similar (or more extreme) in native/introduced habitats. Notably,* L. fortunei *inhabits heavily polluted areas, as well as water with low calcium (3-4mg/L) (Ricciardi 1998).*
- L. fortunei *can exist in areas unsuitable for similar species. In South America, populations of golden mussels thrive in areas that contain numerous pollutants (Zn, Cr, Cu, PCBs) at concentrations above what is considered hazardous for aquatic life (Karatayev et al. 2007).*

B9) How abundant are habitats suitable for the survival, development, and reproduction of this species in the Great Lakes area (e.g., those with adequate depth, substrate, light, temperature, oxygen)?

Abundant (Suitable habitats can be easily found and readily available)	9
Somewhat abundant (Suitable habitats can be easily found but are in high demand by species already present)	6
Somewhat scarce (Suitable habitats can be found occasionally)	3
Scarce (Suitable habitats are rarely found)	0
Unknown	U
	8

- Limnoperna fortunei *reproduction was triggered at 17°C in a study performed by Nakano et al. (2011).*
- L. fortunei *has a depth range of a few centimeters to 10m (Karatayev et al. 2010).*
- L. fortunei *attaches well to hard substrate (including of biological origin), macrophytes and reeds (Karatayev et al. 2007), and plastic bottles (Karatayev et al. 2010).*
- Limnoperna fortunei *attaches minimally to soft substrate, but can attach if the substrate is sufficiently compacted (Boltovskoy et al. 2006).*

B10) How likely is this species to adapt to or to benefit from the predicted effects of climate change on the Great Lakes freshwater ecosystems (e.g., warmer water temperatures, shorter duration of ice cover, altered streamflow patterns, increased salinization)?

Likely (Most of the effects described above make the Great Lakes a better environment for establishment and spread of this species, OR this species could easily adapt to these changes due to its wide environmental tolerances)	9
Somewhat likely (Several of the effects described above could make the Great Lakes a better environment for establishment and spread of this species)	6
Somewhat unlikely (Few of the effects described above would make the Great Lakes a better environment for establishment and spread of this species)	3
Unlikely (Most of the effects described above would have no effect on establishment and spread of this species or would make the environment of the Great Lakes unsuitable)	0
Unknown	U
	9

- Limnoperna fortunei *can withstand very warm temperatures, as well as low levels of salinization.*

- L. fortunei *populations in the Río de la Plata Estuary have been reported at a salinity of 13.7% (Karatayev et al. 2007).*
- *In the Paraná River,* L. fortunei *persists at temperatures as high as 32-33°C (Karatayev et al. 2007).*

B11) How likely is this species to find an appropriate food source (prey or vegetation in the case of predators and herbivores, or sufficient light or nutrients in the case of autotrophs)?

Likely (All possible nutritive food items—including species in the Great Lakes that may be considered potential food items—are highly abundant and/or easily found)	9
Somewhat likely (Some nutritive food items—including species in the Great Lakes that may be considered potential food items—are abundant and/or search time is low to moderate)	6
Somewhat unlikely (Few nutritive food items—including species in the Great Lakes that may be considered potential food items—are abundant and/or search time is moderate to high)	3
Unlikely (All possible nutritive food items—including species in the Great Lakes that may be considered potential food items—are relatively scarce and/or search time is high)	0
Unknown	U
	9

- *The golden mussel consumes a variety of phytoplankton and zooplankton (Frau et al. 2013, Rojas Molina et al. 2012, Rojas Molina et al. 2010).. Adults do well despite low food availability (Oliveira et al. 2010).*
- *Invasive golden mussel populations in Brazil prey on juvenile Daphniidae and Calocalanidae micro crustaceans and Vaginicolidae and Centropyxidae protozoa (Bertao et al. 2021).*
- *Through an isotope analysis, particulate organic matter was identified as the primary food source for* L. fortunei, *plankton and sediment organic matter were also major dietary components* (Zhang et al. 2017)

B12) Does this species require another species for critical stages in its life cycle such as growth (e.g., root symbionts), reproduction (e.g., pollinators, egg incubators), spread (e.g., seed dispersers), or transmission (e.g., vectors)?

Yes, and the critical species (or one that may provide a similar function) is common in the Great Lakes and can be easily found in environments suitable for the species being assessed; OR, No, there is no critical species required by the species being assessed	9
Yes, and the critical species (or one that may provide a similar function) is moderately abundant and relatively easily found in particular parts of the Great Lakes	6
Yes, and the critical species (or one that may provide a similar function) is relatively rare in the Great Lakes AND/OR can only be found occasionally in environments suitable for the species being assessed	3
Yes, and the critical species (or one that may provide a similar function) is not present in the Great Lakes but is likely to be introduced	0
Yes, but the critical species (or one that may provide a similar function) is not present in the Great Lakes and is not likely to be introduced	-80% total points (at end)
Unknown	U
	9

- *At the time of this review,* L. fortunei *does not have any known critical species required for establishment.*

B13) How likely is the establishment of this species to be aided by the establishment and spread of another species already in the Great Lakes?

Likely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established and spread in the Great Lakes, AND available literature predicts this previous invader might promote the establishment of this species, AND/OR there have been cases reported of this species aiding the establishment of this species in other areas)	9
Somewhat likely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established and spread in the Great Lakes)	6
Somewhat unlikely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established in the Great Lakes BUT it is still confined to a small area of the Lakes and the likelihood of encounter with this species assessed is hard to predict)	3
Unlikely (A non-indigenous species to the Great Lakes that facilitates the development of this species has not been established in the Great Lakes)	0
Unknown	U
	8

- L. fortunei *can settle on freshwater invertebrates and mussels with hard shells. In Asia and South America,* L. fortunei *has been observed on bivalves, gastropods, and crustaceans (Darrigran 2002; Karatayev et al. 2010; Rojas Molina and Williner 2013). The abundance of invasive dreissenid mussels in the Great Lakes region would provide suitable biogenic substrate and potentially facilitate the establishment and spread of the golden mussel.*

B14) How likely is establishment of this species to be prevented by the herbivory, predation, or parasitism of a natural enemy that is already present in the Great Lakes and may preferentially target this species?

Likely (The ability of the natural enemy to prevent the establishment of this species in introduced ranges or limiting populations of this species in native ranges is well documented in the literature AND this natural enemy is abundant and widespread in the Great Lakes)	-80% total points (at end)
Somewhat likely (The ability of the natural enemy to prevent the establishment of this species in introduced ranges or limiting populations of this species in native ranges is suggested in the literature OR this natural enemy has limited distribution in the Great Lakes)	-60% total points (at end)
Somewhat unlikely (There are few cases reported of such a natural enemy preventing the establishment of this species in introduced ranges or limiting populations of this species in native ranges OR this natural enemy has low abundance in the Great Lakes)	-10% total points (at end)
Unlikely (Such a natural enemy is particularly rare or is not present in the Great Lakes)	0
Unknown	U
	0

- *While* L. fortunei *is preyed upon by fish, there is no evidence of prevention or mitigation of colonization in other areas (Molloy et al. 1997, Sylvester et al. 2005).*

- *An exclusion experiment in South America suggests that fish predation reduces golden mussel maximum shell length by 40% and density by 70%, but predation did not prevent establishment (Gonzalez-Bergonzoni et al. 2020).*
- Limnoperna fortunei *promotes the growth of the toxic cyanobacteria* Microcystis*. This bacteria is highly toxic to* L. fortunei*'s larvae and limits recruitment. However, larval densities do not decline in the reproductive season following cyanobacterial blooms, suggestive of adult* L. fortunei *immunity to cyanotoxins or behavioral strategies to cope with these toxins (Boltovskoy 2017, Boltovskoy et al. 2021).*

PROPAGULE PRESSURE

B15) On average, how large and frequent are inoculations (introduction events) from the potential vectors identified in Section A for this species? (What is the total number of individuals introduced?)

Frequent, large inocula	9
Frequent, moderate inocula	6
Frequent, small inocula OR infrequent, large inocula	3
Infrequent, small or moderate inocula	0
Unknown	U
	0

- *With little traffic from South America or Asia into the Great Lakes, the inoculation potential would be small.*

HISTORY OF INVASION AND SPREAD

B16) How extensively has this species established reproducing populations in areas outside its native range as a direct or indirect result of human activities?

Very extensively (many invasive populations of this species have been reported in areas widely distributed from the native range)	9
Extensively (some invasive populations of this species have been reported in areas widely distributed from the native range)	6
Somewhat extensively (few invasive populations of this species have been reported in areas widely distributed from the native range OR all invasive populations are in close proximity to each other)	3
Not extensively (no invasive populations of this species have been reported)	0
Unknown	U
	6

- Limnoperna fortunei *was introduced into Hong Kong in 1965 and into Taiwan, Japan, and Argentina around 1990, in the latter case most probably through the Rio de la Plata estuary via ship's ballast water (Boltovskoy et al. 2009). It has since colonized practically the entire Río de la Plata Catchment, including parts of Bolivia, Paraguay, Uruguay and Brazil (Paolucci et al. 2010).*

B17) How rapidly has this species spread by natural means or by human activities once introduced to other locations?

Rapidly (This species has a history of rapid spread in introduced ranges)	9
Somewhat rapidly (This species has a history of moderately rapid spread in introduced ranges)	6
Somewhat slowly (This species has a history of moderately slow spread in its introduced ranges)	3
Slowly (This species has a history of slow to no spread in its introduced ranges)	0
Unknown	U
	9

- Limnoperna fortunei *was introduced into Argentina around 1990, and by the early 2000s L. fortunei had colonized practically all of the Rıo de la Plata watershed extending to Paraguay, Brazil, Uruguay, and Bolivia, and swiftly became the sole dominant organism on hard substrates (Boltovskoy et al. 2006).*
- *Its spread into South America has been notably fast (~400 km/y), covering Argentina, Bolivia, Brazil, Paraguay, and Uruguay by 2016 (Boltovskoy 2017).*

B18) Are there any existing control measures in the Great Lakes set to prevent the establishment and/or spread of this species?

Yes, and they are likely to prevent establishment or spread of the species. (There are no reported cases of this species adapting or avoiding current measures. These measures are highly effective in preventing the establishment and spread of this species)	-90% total points (at end)
Yes, and they are moderately likely to prevent establishment or spread of the species. (There are few reported cases of this species adapting or avoiding current measures used to control its establishment and spread)	-50% total points (at end)
Yes, but they are unlikely to prevent establishment or spread of the species. (There are many reported cases of this species adapting or avoiding current measures used to control its establishment and spread)	-20% total points (at end)
No control methods have been set to prevent the establishment and/or spread of this species	0
Unknown	U
	-20%

- *Apart from regulatory restrictions, there are no existing control measures in the Great Lakes to prevent the establishment of* Limnoperna fortunei. *However, this species is being actively watched for in surveillance and monitoring efforts, which will limit its establishment potential.*

Section B Scores				
Points	**Probability of Establishment**	**Total Points (pre-adjustment)**		116
>100	High	**Adjustments**		
		Critical species	A (116-0%)	116
51-99	Moderate	**Natural enemy**	B (116-0%)	116
		Control measures	C (116-20%)	93
0-50	Low	**Total Points (post-adjustment)**		**93**
		Probability of Establishment		**Moderate**
# of questions answered as "unable to determine"	**Confidence Level**			
0-1	High	**Total # of questions unknown**		**0**
2-5	Moderate			
6-9	Low	**Confidence Level**		**High**

<u>Qualitative Statements for GLANSIS Fact Sheet:</u>

Limnoperna fortunei has a moderate probability of establishment if introduced to the Great Lakes (Confidence level: High).

Comments:

Status: Not established in North America, including the Great Lakes

Limnoperna fortunei can survive salinity shocks and changes, allowing its transportation into the Great Lakes and the ability to acclimate to salinities around the Great Lakes. This species is already found in regions with climates similar to the Great Lakes, and shows the ability to survive overwinter temperatures as low as 0°C. *L. fortunei* consumes a variety of phytoplankton and zooplankton making it able to adjust its diet based on prey availability in the Great Lakes if introduced.

The native and introduced ranges of *L. fortunei* include extremes of pollution, water temperature, pH, and nutrient levels, making it able to adapt to the many microhabitats throughout the Great Lakes. *Limnoperna fortunei* attaches well to hard substrate (including of biological origin),as well as macrophytes and reeds (Karatayev et al. 2007) and plastic bottles (Karatayev et al. 2010). *Limnoperna fortunei* attaches minimally to soft substrate, but can attach if the substrate is sufficiently compacted (Boltovskoy et al. 2006).

SECTION C: POTENTIAL IMPACT ASSESSMENT RESULTS

Environmental: High
Socio-Economic: High
Beneficial: High

Comments:

***Limnoperna fortunei* has the potential for high environmental impact if introduced to the Great Lakes.**

Limnoperna fortunei reach large densities (5,000-250,000 individuals/m2 on hard substrate, and 90-2,000 individuals/m2 on soft substrate (Frau et al. 2012); shifting productivity in the nutrient cycle from the pelagic zone to the benthic zone. This species filters water quickly, causing a reduction in primary production occurring within the water column. Nutrient concentrations and proportion are shifted to promote aggregation of solitary *Microcystis* spp. cells; this favors blooms of the noxious cyanobacteria (Cataldo et al. 2012). *L. fortunei* bioaccumulates heavy metals and pesticides and may facilitate the transfer of these substances to existing fauna (Besen and Marengoni 2021, Sene et al. 2021).

Limnoperna fortunei has been shown to physically foul living unionids and naiads through settlement on their shells. This impairs movement and restricts their ability to open their valves, thus depriving them of food and oxygen (Darrigan and Damborenea 2005, Weigand 2019). This species also significantly impacts benthic fauna, with changes in invertebrate and plankton abundance well-documented (Rojas Molina et al. 2012, Frau et al. 2012, Bertao et al. 2021, Silva et al. 2021). *L. fortunei* is an ecosystem engineer that significantly alters invaded habitats by altering habitat complexity, sedimentation, and accelerating eutrophication (Burlakova et al. 2012, Tokumon et al. 2018, Gattas et al. 2020)

***Limnoperna fortunei* has the potential for high socio-economic impact if introduced to the Great Lakes.**

L. fortunei modifies nutrient concentrations and proportions, and promotes aggregation of solitary *Microcystis* spp. cells into colonies; both these effects can favor blooms of this often noxious cyanobacteria (Cataldo et al. 2012a). Gazulha et al. (2012) found that while single cells of cyanobacteria were accepted, filamentous and colonial cyanobacteria were rejected as pseudofeces.*L. fortunei* has been shown to increase the growth of *Microcystis* spp. through the alteration of P:N ratios, selective grazing, competitor exclusion, and increasing nutrient supply (Boltovskoy 2017, Silva and Giani 2018, Gangi et al. 2020).

Limnoperna fortunei can clog fouling water intake sieves and filters, pipes, heat exchangers, and condensers. This species has become a common difficulty in industrial and power plants that use raw water, chiefly for cooling purposes (Boltovskoy et al. 2009, Cataldo et al. 2003, Goto 2002). Their biofouling leads to increased abrasive wear and significant economic costs for their cleaning and maintenance (Yao et al. 2017, de Castro et al. 2019, Boltovskoy et al. 2022). *L. fortunei* biofouls 38% of the hydropower plants in Brazil and is responsible for estimated annual losses of 120 million dollars (US$) (SPIC Brasil 2021).

The current solutions implemented in the Great Lakes to control dreissenid mussels are expected to be effective against *Limnoperna fortunei*. As such, there are many control measures in place to mitigate socio-economic impact from golden mussel invasion. The potential for increased impacts exists if *L. fortunei* is resistant to existing control measures or able to occupy areas where dreissenids are excluded due to their increased physiological tolerances.

***Limnoperna fortunei* has the potential for high beneficial impact if introduced to the Great Lakes.**

Limnoperna fortunei has had similar impacts to *Dreissena polymorpha*, i.e., has led to a dramatic shift in the benthic trophic structure and a homogenization of freshwater communities, regardless of original substrate and community structure (Burlakova et al. 2012, Darrigran and Damborenea 2011, Sardiña et al. 2011). Also, this species has led to an increase in water transparency, a decrease in suspended matter, chlorophyll a, and primary production (Boltovskoy et al. 2009). *L. fortunei* promotes glyphosate degradation and reduces the concentration of herbicides in freshwater systems (Di Fiori et al. 2012, Vargas et al. 2019). The structure of functional feeding groups in the new community, including invasive bivalves, is overwhelmingly dominated by collectors-filterers (Burlakova et al. 2012). Predictive models for the Lake Erie food web suggest *Limnoperna fortunei* will positively impact piscivores, most omnivores (except juvenile yellow perch), and most benthic invertebrates (except dreissenids), negative effects are predicted for planktivores and plankton (Zhang et al. 2019).

Introduction of *Limnoperna fortunei* into habitats outside of its native range provides a new food resource to be exploited by the fish community. This species has become an important food resource for native fish, including those of commercial value (Paolucci et al. 2017, de Ávila-Simas et al. 2019, Gonzalez-Bergonzoni et al. 2020). An increase in mussel densities have been associated with a threefold increase in Argentina's freshwater fish landings in the Rio do la Plata basin after 1995, due to its status as a new, abundant food source (Boltovskoy et al. 2006). Positive impacts of *L. fortunei* are not limited to fishes that directly consume mussels, but there are also indirect positive effects on ichthyophagous and detritivorous fish species.

Parallels with the *Dreissena polymorpha* highlight several important differences, including *Limnoperna fortunei*'s higher resistance to anoxia, pollution, pH, and high temperatures, longer reproduction periods and lower calcium requirements (Karatayev et al. 2007), suggesting that, should *Limnoperna fortunei* reach North America or Europe, it will become an even more aggressive invader, especially in regions dominated by acidic, soft and contaminated waters (Boltovskoy et al. 2009).

POTENTIAL ENVIRONMENTAL IMPACT

C1) Does the species pose some hazard or threat to the health of native species (e.g., it magnifies toxin levels; is poisonous; is a pathogen, parasite, or a vector of either)?

Yes, and it has impacted threatened/endangered species, resulted in the reduction or extinction of one or more native populations, affects multiple species, or is a reportable disease	6 √
Yes, but negative consequences have been small (e.g., limited number of infected individuals, limited pathogen transmissibility, mild effects on populations and ecosystems)	1
Not significantly	0
Unknown	U

- *In a mesocosm experiment, cell density, proportion of colonial cells and colony size of* Microcystis *spp. increased in all enclosures, but these increases were dramatically (and very significantly) higher in enclosures with 100 and, especially, with 300 golden mussels, than in the controls. Enclosure density reflected realistic densities. The results indicated that* L. fortunei *modifies nutrient concentrations and proportions, and promotes aggregation of solitary* Microcystis *spp. cells into colonies; both these effects can favor blooms of this often noxious cyanobacteria (Cataldo et al. 2012).*
- *Mechanisms through* L. fortunei *promotes cyanobacteria blooms may include: (i) modification of nutrient concentrations and the N : P ratio, (ii) selective grazing of solitary* Microcystis *spp. cells over colonial ones and (iii) production of chemical cues that trigger the formation of colonies (Boltovskoy et al. 2009).*
- *Gazulha et al. (2012) found that while single cells of cyanobacteria were accepted by* L. fortunei*, filamentous and colonial cyanobacteria were rejected as pseudofeces, a preference that could enhance blooms.*
- L. fortunei *may also biomagnify contaminants, i.e., the pseudofeces that primarily consist of rejected contaminants are consumed by amphipods (Sardiña et al. 2011).*
- *Another potential threat posed by this invader was reported by Ogawa et al. (2004). The authors identified widespread parasitic infections by bucephalid trematodes in several cyprinid fishes from the Uji river (Japan), suggesting that the infections started with the accidental introduction of infested first intermediate hosts (*Limnoperna fortunei*) (Ogawa et al. 2004).*
- L. fortunei *specimens attached to net cages in a Brazilian reservoir bioaccumulate high levels of heavy metals (Cd, Pb, Al, Zn, and Cr). Their high bioaccumulation rates may facilitate the transfer of heavy metals to existing fauna (Besen and Marengoni 2021).*
- *This species bioaccumulates pesticides, organochlorines and organophosphates in particular, and may facilitate widespread food-web contamination in freshwater ecosystems (Sene et al. 2021).*
- *In South America,* L. fortunei *invasion has led to increased biomagnification of contaminants in the sediment, this has led to population declines for ostracods and nematodes (Boltovskoy et al. 2015).*

C2) Does it out-compete native species for available resources (e.g., habitat, food, nutrients, light)?

Yes, and it has resulted in significant adverse effects (e.g., impacted threatened/endangered species or caused critical reduction, extinction, behavioral changes including modified spawning behavior) on one or more native populations	6 √
Yes, and it has caused some noticeable stress to (e.g., decrease in growth, survival, fecundity) or decline of at least one native population	1
Not significantly	0
Unknown	U

- Limnoperna fortunei *may outcompete/overgrow the native South American gastropods,* Chilina fluminae *and* Biomphalaria straminea*, as they became rare and eventually disappeared from study sites (Spaccesi and Capitulo 2012).*
- *The golden mussel negatively effects burrowing invertebrates and unionids (Karatayev et al. 2010).*
- L. fortunei *negatively impacts zooplankton, and part of zooplankton decline may be due to starvation (i.e., mussel outcompetes zooplankton for food resources) (Rojas Molina et al. 2012).*
- *The shell production by* Limnoperna fortunei *decreases flow velocity, increases shear strength and increases sedimentation. These ecosystem engineering effects lead to spatial interference competition with native bivalves, reed roots, etc. (Darrigan and Damborenea 2011).*
- *In South America,* L. fortunei *attaches to, and displaces, native naiads (Hyriidae and Mycetopodidae) by inhibiting their ability to open or close their valves (Darrigan and Damborenea 2005).*
- *L. fortunei has been shown to physically foul living unionids through settlement on their shells. This impairs movement and restricts their ability to open their valves, thus depriving them of food and oxygen (Weigand 2019).*
- L. fortunei *has been found to attach to a native crab,* Trichodactylus borellianus*, which leads to higher energetic costs for locomotion and reduces ability to escape predators (Rojas Molina and Williner 2013).*

C3) Does it alter predator-prey relationships?

Yes, and it has resulted in significant adverse effects (e.g., impacted threatened/endangered species, caused significant reduction or extinction of one or more native populations, creation of a dead end or any other significant alteration in the food web)	6 √
Yes, and it has resulted in some noticeable stress to (e.g., decrease in growth, survival, fecundity) or decline of at least one native population AND/OR Yes, and it has resulted in some alteration of the food web structure or processes, the effects of which have not been widespread or severe	1
Not significantly	0
Unknown	U

- Limnoperna fortunei *has increased food availability in the benthic zone which subsequently led to increased invertebrate density and biomass. Burlakova et al. (2012) report a density increase of 1.9 to 22 times higher and biomass increase of 1.7 to 19 times higher white Karatayev et al. (2010) report a threefold increase in benthic invertebrate density and biomass, these changes have the potential to alter the food web.*
- L. fortunei *has homogenized benthic communities (Darrigran and Damborenea 2011, Sardiña et al. 2011); e.g., in one study 99.9% of community biomass consisted of the filtering collector trophic group (Burlakova et al. 2012)*
- L. fortunei *has shunted the dominant nutrient cycling from the pelagic to the benthic zone (Cataldo et al. 2012, Darrigran and Damborenea 2011, Rojas Molina et al. 2012).*
- *The golden mussel has been found to significantly reduce phytoplankton densities (>60%). Also, it has changed composition of the algae assemblages, most notably an increase in the flagellate group, relative to the diatom, single celled and colonial groups (Frau et al. 2012).*
- *Phytoplankton decreased 55-90% in mesocosm experiment with golden mussel (Rojas Molina et al. 2012).*
- L. fortunei *filters water substantially faster than* D. polymorpha *(Karatayev et al. 2010).*
- *The golden mussel consumes zooplankton:*
    - *Copepodites experienced 30% mortality in golden mussel treatments, compared to controls (Rojas Molina et al. 2013).*
    - *Gut content analysis found that microcrustaceans constitute about 67% of golden mussel diet biomass (Rojas Molina et al. 2010).*
    - *Significant declines in rotifers and cladocerans in mesocosm experiments with golden mussels (Rojas Molina et al. 2012)*

- L. fortunei *selectively grazes Chlorophyceae, Euglenophyceae and Chrysophyceae phytoplankton. Additionally, L. fortunei avoids grazing on filamentous, colonial Cyanobacteria, and Bacillariophyceae (Frau 2022).*

- L. fortunei *can reach densities of 5000-250,000 ind/m2 on hard substrate, and 90-2000 ind/m2 on soft substrate (Frau et al. 2012).*
- *The accumulation of* L. fortunei *shells and enrichment of the habitat through their pseudofeces and biodeposits favors an increase in the abundance of macroinvertebrates from the Chironomidae family. The enrichment also alters benthic fauna abundance and can cause a biological imbalance in the environment (Bertao et al. 2021).*
- *In field experiments,* Limnoperna fortunei *reduced densities of benthic macroinvertebrates, scrapers, dipterans, and gastropods (Silva et al. 2021).*
- *Predictive models for the Lake Erie food web suggest* Limnoperna fortunei *will positively impact piscivores, most omnivores (except juvenile yellow perch), and most benthic invertebrates (except dreissenids). Negative effects are predicted for planktivores and plankton (Zhang et al. 2019).*

C4) Has it affected any native populations genetically (e.g., through hybridization, selective pressure, introgression)?

Yes, and it has caused a loss or alteration of genes that may be irreversible or has led to the decline of one or more native species (or added pressure to threatened/endangered species)	6
Yes, some genetic effects have been observed, but consequences have been limited to the individual level	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that* Limnoperna fortunei *affects native populations genetically. There are no congeneric species native to the Great Lakes.*

C5) Does it negatively affect water quality (e.g., increased turbidity or clarity, altered nutrient, oxygen, or other chemical levels/cycles)?

Yes, and it has had a widespread, long-term, or severe negative effect on water quality AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6 √
Yes, it has affected water quality to some extent, but the alterations and resulting adverse effects have been limited or inconsistent (as compared with above statement)	1
Not significantly	0
Unknown	U

- *An increase in water transparency, and a decrease in suspended matter, chlorophyll a, and primary production has been observed after golden mussel establishment (Boltovskoy et al. 2009).*
- *A decrease in turbidity and an increase in dissolved nitrogen has been found when golden mussels are present (Rojas Molina et al. 2012).*
- L. fortunei *has been shown to increase the growth of* Microcystis *spp., bloom forming cyanobacteria, through the alteration of P:N ratios, selective grazing, competitor exclusion, and increasing nutrient supply (Boltovskoy et al. 2017, Silva and Giani 2018, Gangi et al. 2020).*
- *Microcosm experiments found that* L. fortunei *mussel colonies, dead mussel colonies in particular, fostered the growth of pathogenic bacteria (Jiahao 2022).*
- *In microcosm experiments,* L. fortunei *metabolized glyphosate and accelerated eutrophication (Gattas et al. 2020).*
- *In microcosm experiments,* L. fortunei *changed the nutrient content in the water column in such a way that could favor the development of certain phytoplankton groups over others. This could also inadvertently select for faster growing phytoplankton and increase their abundance (De Stefano et al. 2018).*

C6) Does it alter physical components of the ecosystem in some way (e.g., facilitated erosion/siltation, altered hydrology, altered macrophyte/phytoplankton communities, physical or chemical changes to substrate)?

Yes, and it has had a widespread, long term, or severe negative effect on the physical ecosystem AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6 √
Yes, it has affected the physical ecosystem to some extent, but the alterations and resulting adverse effects have been mild	1
Not significantly	0
Unknown	U

- Limnoperna fortunei *are habitat engineers that provide shelter to other species when density is high and colonies are formed (Weigand 2019).*
- *Increased habitat complexity can lead to significant (e.g., threefold) increase in community taxonomic richness. Golden mussel shells increase surface area for settling organisms, and also provide refuges from predation and physical stressors (Burlakova et al. 2012, Darrigran et al. 1998, Darrigran and Damborenea 2011, Spaccesi and Capitulo 2012). This species transforms sand or mostly bare sediment into reef-like druses (Burlakova et al. 2012).*
- *In the Río de la Plata estuary,* L. fortunei *doubles the rate of sedimentation for suspended particles and increases their organic matter and nitrogen contents. These changes are expected to increase accretion rates and alter the communities that occupy the sediments (Tokumon et al. 2018).*
- *In South America,* L. fortunei *colonization has led to the eutrophication of substrates (Karatayev et al. 2010).*
- L. fortunei *indirectly promotes the growth of periphyton and aquatic macrophytes, this occurs as a result of its impacts on nutrient cycling and water clarity (Boltovskoy et al. 2015). A reservoir in South America was observed to have no significant macrophyte populations prior to golden mussel colonization. Two years after* L. fortunei *colonized the reservoir, the macrophyte* Elodea callitrichoides *abundance increased to the point that it was a significant feature of the waterbody with large beds along the coastlines (Boltovskoy et al. 2009).*
- L. fortunei *may not increase habitat complexity given the abundance of dreissenid mussels in the Great Lakes but may have a substantial impact in areas where dreissenids are currently excluded due to its tolerance of pH, pollution, and oxygen deprivation.*

Environmental Impact Total	**30**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

POTENTIAL SOCIO-ECONOMIC IMPACT

C7) Does the species pose some hazard or threat to human health (e.g., it magnifies toxin levels, is poisonous, a virus, bacteria, parasite, or a vector of one)?

Yes, significant effects on human health have already been observed	6
Yes, but negative consequences have not been widespread, long lasting, or severe	1 √
Not significantly	0
Unknown	U

- *In a mesocosm experiment, cell density, proportion of colonial cells and colony size of* Microcystis *spp. increased in all enclosures, but these increases were dramatically (and very significantly) higher in enclosures with 100 and, especially, with 300 golden mussels, than in the controls. Enclosure density reflected realistic densities. The results indicate that* L. fortunei *modifies nutrient concentrations and proportions, and promotes aggregation of solitary* Microcystis *spp. cells into colonies; both these effects can favor blooms of this often noxious cyanobacteria (Cataldo et al. 2012).*
- *Gazulha et al. (2012) found that while single cells of cyanobacteria were accepted, filamentous and colonial cyanobacteria were rejected by* L. fortunei *as pseudofeces, a preference that could enhance blooms.*
- L. fortunei *has been shown to increase the growth of* Microcystis *spp. through the alteration of P:N ratios, selective grazing, competitor exclusion, and increasing nutrient supply (Boltovskoy et al. 2017, Silva and Giani 2018, Gangi et al. 2020).*

C8) Does it cause damage to infrastructure (e.g., water intakes, pipes, or any other industrial or recreational infrastructure)?

Yes, it is known to cause significant damage	6 √
Yes, but the costs have been small and are largely reparable or preventable	1
Not significantly	0
Unknown	U

- *L. fortunei can clog or foul water intake sieves and filters, pipes, heat exchangers, and condensers has become a common difficulty in industrial and power plants that use raw water, chiefly for cooling purposes (Boltovskoy et al. 2009, Cataldo et al. 2003, Goto 2002,).*
- *Grit chambers and flocculators have been found heavily clogged with sediment, broken shells and tissue material from golden mussel (Crosier and Molloy 2004 , Goto 2002, Magara et al. 2001). This increases operational costs (complete shutdown of plant; clogging of mussels, shell material, and sediment may need to be manually removed) (Crosier and Molloy 2004, Goto 2002).*
- L. fortunei *is a significant problem for water transfer facilities, industrial installations, and power plants. Their biofouling leads to significant economic costs for their cleaning and maintenance (Boltovskoy et al. 2022).*
- L. fortunei *shells become embedded in hydroelectric plants and increase abrasive wear as they pass through the system (de Castro et al. 2019).*
- *Concrete structures fouled by* L. fortunei *experience increased deterioration and wear (Yao et al. 2017).*
- L. fortunei *biofouls 38% of the hydropower plants in Brazil and is responsible for estimated annual losses of 120 million dollars (US$) (SPIC Brasil 2021).*

C9) Does it negatively affect water quality (i.e. in terms of being less suitable for human use)?

Yes, it has significantly affected water quality, and is costly or difficult to reverse	6
Yes, but the effects are negligible and/or easily reversed	1 √

Not significantly	0
Unknown	U

- *Decaying Golden Mussels pollute drinking water systems (Crosier and Molloy 2004).*

C10) Does it negatively affect any markets or economic sectors (e.g., commercial fisheries, aquaculture, agriculture)?

Yes, it has caused significant damage to one or more markets or economic sectors	6
Some damage to markets or sectors has been observed, but negative consequences have been small	1 √
Not significantly	0
Unknown	U

- Limnoperna fortunei *invasion increases production costs for tilapia fish farming in Brazil. Reported added costs were R$ 0.45 kg/fish for large-size fish farms and R$ 1.00 kg/fish for small- and medium-size fish farms (da Costa et al. 2018).*

C11) Does it inhibit recreational activities and/or associated tourism (e.g., through frequent water closures, equipment damage, decline of recreational species)?

Yes, it has caused widespread, frequent, or otherwise expensive inhibition of recreation and tourism	6
Yes, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species inhibits recreation or tourism.*

C12) Does it diminish the perceived aesthetic or natural value of the areas it inhabits?

Yes, the species has received significant attention from the media/public, significantly diminished the natural or cultural character of the area, or significantly reduced the area's value for future generations	6
Yes, but negative consequences have been small	1 √
Not significantly	0
Unknown	U

- *Decaying dead golden mussels emit a noxious odor (Crosier and Molloy 2004).*
- *In microcosm experiments,* L. fortunei *reduced glyphosate effectiveness, altered nutrient concentrations, and increased eutrophication in the water body. The resulting overabundance of algae decreases recreational value (Gattas et al. 2020).*

Socio-Economic Impact Total	**10**
Total Unknowns (U)	**0**

Scoring

Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

POTENTIAL BENEFICIAL EFFECT

C13) Does it act as a biological control agent for aquatic weeds or other harmful nonindigenous organisms?

Yes, it has succeeded significantly as a control agent	6
Yes, it has had some success as a control agent, but may be inconsistent or lack a desired level of effectiveness	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species acts as a biological control agent.*

C14) Is it commercially valuable (e.g., for fisheries, aquaculture, agriculture, bait, ornamental trade)?

Yes, it is economically important to at least one of these industries	6
Yes, but its economic contribution is small	1
Not significantly	0 √
Unknown	U

- L. fortunei *shells are composed of calcium carbonate and may be used as mulch, soil amendment, or fertilizer in agriculture. Mussels can also be harvested as food for livestock and farmed fish. However, there are no reports of commercial harvest for* L. fortunei *and their patchy distribution may make harvest not viable from an economic standpoint (Burlakova et al. 2023).*

C15) Is it recreationally valuable (e.g., for sport or leisurely fishing, as a pet, or for any other personal activity)?

Yes, it is commonly employed recreationally and has some perceived value for local communities and/or tourism	6
It is sometimes employed recreationally, but adds little value to local communities or tourism	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species is recreationally valuable.*

C16) Does the species have some medicinal or research value (i.e. outside of research geared towards its control)?

Yes, it has significant medicinal or research value	6
It has some medicinal or research value, but is not of high priority OR It is potentially important to medicine or research and is currently being or scheduled to be studied	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species has any medicinal or research value.*

C17) Does the species remove toxins or pollutants from the water or otherwise increase water quality?

Yes, it reduces water treatment costs or has a significant positive impact for the health of humans and/or native species	6
Yes, but positive impact for humans or native species is considered negligible	1 √
Not significantly	0
Unknown	U

- *In experiments performed by Di Fiori et al. (2012) glyphosate decreased by 40% under large mussel presence, and was reduced by 25% in empty shell treatments. The authors believe that part of the herbicide that disappeared from the water column was adsorbed in the golden mussel valve surface, while another was mineralized by microbial communities in shells' biofilm. Glyphosate is an herbicide with negative impacts on aquatic filtering organisms and sediment feeders. However, the golden mussel may speed up the bioavailability of the phosphorus in glyphosate, leading to eutrophication, but this is expected to be outweighed by the remediation of the herbicide (Di Fiori et al. 2012).*
- L. fortunei *promotes glyphosate degradation and reduces the concentration of herbicides in freshwater systems (Vargas et al. 2019).*
- L. fortunei *beds or shells can be utilized to remove contaminants such as pesticides and pharmaceuticals from water and wastewater, but have not yet been utilized outside of experimental environments (Burlakova et al. 2023).*

C18) Does the species have a positive ecological impact outside of biological control (e.g., increases the growth or reproduction rates of other species, fills an important gap in the food web, supports the survival of a species that is threatened, endangered species, or commercially valuable)?

Yes, it significantly contributes to the ecosystem in one or more of these ways	6 √
Yes, it provides some positive contribution to the ecosystem, but is not vital	1
Not significantly	0
Unknown	U

- *An increase in golden mussel densities have been associated with a threefold increase in Argentina's freshwater fish landings in the Río de la Plata basin after 1995, due to its status as a new, abundant food source (Boltovskoy et al. 2006). The Río de la Plata system is otherwise poor in plankton (Boltovskoy et al. 2006).*
- *In a study of 15 fish taxa, 11 had veligers of* L. fortunei *in their gut (several of these fish species are economically and environmentally important). Fish larvae with empty guts represented 60% (San Nicolás) to 72% (Paraná River) of the total number of fish. Proportions of feeding fish larvae with* L. fortunei *veligers in their guts varied between 20% (San Nicolás) and 56% (Paraná River); in 15% of the guts analyzed,* L. fortunei *was the only food item recorded. For those specimens that had consumed* L. fortunei *larvae and any other food item,* L. fortunei *was the most important item in 55% (Paraná River) to 71% (San Nicolás) of the animals in terms of biomass. Consumption of* L. fortunei *veligers decreased as fish larvae grew and were able to capture more difficult prey (Paolucci et al. 2007).*
- *Feeding on veligers of* L. fortunei *significantly enhanced the growth of* P. lineatus *larvae and supported the idea that this new and abundant resource is selectively preyed upon by* P. lineatus *during its larval stage. Higher growth rates may have stemmed from the higher energy contents of veligers compared to crustaceans and/or from the lower energy costs of capturing slower prey (Paolucci et al. 2010).*
- *According to Boltovskoy et al. (2006), positive impacts of* L. fortunei *are not limited to fishes that directly consume Golden Mussels, but there are also indirect positive effects on ichthyophagous and detritivorous fish species. Fish larvae, e.g.* Salminus maxillosus*,* Rhaphiodon vulpinus*,* Pseudoplatystoma coruscans*,* Pseudoplatystoma fasciatum*, and metalarvae of other* Pimelodidae *that prey on other larval fish probably benefit from the mussel-enhanced growth rates of their prey.*

- *An observational study in the upper Uruguay River found that 22 fish species consumed* L. fortunei. *Notably, 11 species were not previously known to incorporate* L. fortunei *in their diet (de Ávila-Simas et al. 2019). The golden mussel composed greater than 70% of the feeding index for* Cyprinus carpio, Crenicichla minuano, Crenicichla celidochilus, Crenicichla jurubi, *and* Paraloricaria vetula. *Additionally, results suggest that* C. carpio *feeds exclusively on* L. fortunei.
- *In the lower Uruguay River, 28 out of the 81 species in the fish assemblage consumed* L. fortunei *(Gonzalez-Bergonzoni et al. 2020).*
- L. fortunei *has been used as a biomonitor for the presence of environmental pollutants (Besen and Marengoni 2021).*

Beneficial Effect Total	**7**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

REFERENCES

Andrade, J.T.M., N.I.S. Cordeiro, L.C. Montresor, D.M.R. Luz, R.C.R. Luz, C.B. Martinez, J. Pinheiro, A.P. Paglia, and T. Vidigal. 2018. Effect of temperature on behavior, glycogen content, and mortality in *Limnoperna fortunei* (Dunker, 1857) (Bivalvia: Mytilidae). Journal of Limnology 77(2):189-198. https://doi.org/10.4081/jlimnol.2017.1658.

Angonesi, L.G., N.G. da Rosa, and C.E. Bemvenuti. 2008. Tolerance to salinities shocks of the invasive mussel *Limnoperma fortunei* under experimental conditions. Iheringia Serie Zoologia 98(1):66-69. https://doi.org/10.1590/S0073-47212008000100009.

Ayroza, D., T.J. de Melo, L.P. Faria-Pereira, M.L. Petesse, M.D. Rebelo, C.F. do Carmo, and D.H. Cataldo. 2021. *Limnoperna fortunei* colonization and macrofouling on net cages in a subtropical reservoir (Brazil). Aquaculture Research 52(11):5472-5482. https://doi.org/10.1111/are.15417.

Bertao, A.P.D., R.V.V. Leite, A. Horodesky, M.R. Pie, T.L. Zanin, O.S.M. Netto, and A. Ostrensky. 2021. Ecological interactions between invasive and native fouling species in the reservoir of a hydroelectric plant. Hydrobiologia 848(21):5169-5185. https://doi.org/10.1007/s10750-021-04706-7.

Besen, M.A., and N.G. Marengoni. 2021. Bioaccumulation of metals and evaluation of golden mussels encrusted on different screens of net cages. Boletim Do Instituto De Pesca 47:e624. https://doi.org/10.20950/1678-2305/bip.2021.47.e624.

Boltovskoy, D. 2015. *Limnoperna Fortunei*: The Ecology, Distribution and Control of a Swiftly Spreading Invasive Fouling Mussel. Volume 10. 1st edition. Springer Cham, Switzerland. https://doi.org/10.1007/978-3-319-13494-9.

Boltovskoy, D. 2017. Traits and impacts of invasive species: Myths and evidences from the perspective of introduced freshwater mussels. Aquatic Ecosystem Health & Management 20(4):334-343. https://doi.org/10.1080/14634988.2017.1397483.

Boltovskoy, D., A. Karatayev, L. Burlakova, D. Cataldo, V. Karatayev, F. Sylvester, and A. Mariñelarena. 2009. Significant ecosystem-wide effects of the swiftly spreading invasive freshwater bivalve *Limnoperna fortunei*. Hydrobiologia 636(1):271-284. dx.doi.org/10.1007/s10750-009-9956-9.

Boltovskoy, D., F. Bordet, V. Leites, and D. Cataldo. 2021. Multiannual trends (2004-2019) in the abundance of larvae of the invasive mussel *Limnoperna fortunei* and crustacean zooplankton in a large South American reservoir. Austral Ecology 46(8):1221-1235. https://doi.org/10.1111/aec.13058.

Boltovskoy, D., N. Correa, D. Cataldo, and F. Sylvester. 2006. Dispersion and ecological impact of the invasive freshwater bivalve *Limnoperna fortunei* in the Río de la Plata watershed and beyond. Biological Invasions 8(4):947-963. https://doi.org/10.1007/s10530-005-5107-z.

Boltovskoy, D., R. Guiasu, L. Burlakova, A. Karatayev, M.A. Schlaepfer, and N. Correa. 2022. Misleading estimates of economic impacts of biological invasions. Ambio 51(8):1786-1799. https://doi.org/10.1007/s13280-022-01707-1.

Burlakova, L.E., A.Y. Karatayev, and V.A. Karatayev. 2012. Invasive mussels induce community changes by increasing habitat complexity. Hydrobiologia 685(1):121-134. dx.doi.org/10.1007/s10750-011-0791-4.

Burlakova, L.E., A.Y. Karatayev, D. Boltovskoy, and N.M. Correa. 2023. Ecosystem services provided by the exotic bivalves *Dreissena polymorpha*, *D. rostriformis bugensis*, and *Limnoperna fortunei*. Hydrobiologia 850:2811-2854. https://doi.org/10.1007/s10750-022-04935-4.

Cataldo, D., A. Vinocur, I. O'Farrell, E. Paolucci, V. Leites, and D. Boltovskoy. 2012. The introduced bivalve *Limnoperna fortunei* boosts *Microcystis* growth in Salto Grande reservoir (Argentina): evidence from mesocosm experiments. Hydrobiologia 680(1):25-38. http://link.springer.com/article/10.1007/s10750-011-0897-8.

Cataldo, D., D. Boltovskoy, and M. Pose. 2003. Toxicity of chlorine and three nonoxidizing molluscicides to the pest mussel *Limnoperna fortunei*. Journal (American Water Works Association) 95(1):66-78. https://doi.org/10.1002/j.1551-8833.2003.tb10270.x.

Crosier, D.M., and D.P. Molloy. 2004. Golden Mussel. United States Army Corps of Engineers. http://el.erdc.usace.army.mil/ansrp/limnoperna_fortunei.pdf

da Costa, J.I., M.I.E. Martins, and D. Ayroza. 2018. Impact of control of the golden mussel on the production costs of tilapia bred in net cages. Boletim Do Instituto De Pesca 44(1):110-115. https://doi.org/10.20950/1678-2305.2018.284.

Darrigran, G., and C. Damborenea. 2005. A South American bioinvasion case history: *Limnoperna fortunei* (Dunker, 1857), the golden mussel. American Malacological Bulletin 20(1/2):105-112. http://www.malacologia.com.ar/MALACOLOGIA/PDF/Darrigran_Damborenea_2005.pdf.

Darrigran, G., and C. Damborenea. 2011. Ecosystem engineering impact of *Limnoperna fortunei* in South America. Zoological Science 28(1):1-7. dx.doi.org/10.2108/zsj.28.1.

Darrigran, G., S.M. Martin, B. Gullo, and L. Armendariz. 1998. Macroinvertebrates associated with *Limnoperna fortunei* (Dunker, 1857) (Bivalvia, Mytilidae) in Río de Plata, Argentina. Hydrobiologia 367(1):223-230. https://doi.org/10.1023/A:1003244603854.

De Andrade, J.T.M., N.I.S. Cordeiro, L.C. Montresor, D.M.R. da Luz, E.M.D. Viana, C.B. Martinez, and T. Vidigal. 2021. Tolerance of *Limnoperna fortunei* (Dunker, 1857) (Bivalvia: Mytilidae) to aerial exposure at different temperatures. Hydrobiologia 848(12-13):2993-3001. https://doi.org/10.1007/s10750-020-04191-4.

Deaton, L.E., J.G.S. Derby, N. Subhedar, and M.J. Greenberg. 1989. Osmoregulation and salinity tolerance in two species of bivalve mollusc: *Limnoperna fortunei* and *Mytilopsis leucophaeta*. Journal of Experimental Marine Biology and Ecology 133(1-2):67-79. https://doi.org/10.1016/0022-0981(89)90158-5.

de Avila-Simas, S., D.A. Reynalte-Tataje, and E. Zaniboni. 2019. Fish predators of the golden mussel *Limnoperna fortunei* in different environments in a South American subtropical river. Boletim Do Instituto De Pesca 45(2):e484. https://doi.org/10.20950/1678-2305.2019.45.2.484.

de Castro, A.L.P., R.O.P. Serrano, M.A. Pinto, G. da Silva, L.D.A. Ribeiro, E.M.D. Viana, and C.B. Martinez. 2019. Case study: Abrasive capacity of *Limnoperna fortunei* (golden mussel) shells on the wear of 3 different steel types. Wear 438:202999. https://doi.org/10.1016/j.wear.2019.202999.

De Stefano, L.G., F. Gattas, A. Vinocur, D. Cristos, D. Rojas, D. Cataldo, and H. Pizarro. 2018. Comparative impact of two glyphosate-based formulations in interaction with *Limnoperna fortunei* on freshwater phytoplankton. Ecological Indicators 85:575-584. https://doi.org/10.1016/j.ecolind.2017.11.021.

Di Fiori, E., H. Pizarro, M. dos Santos Afonso, and D. Cataldo. 2012. Impact of the invasive mussel *Limnoperna fortunei* on glyphosate concentration in water. Ecotoxicology and Environmental Safety 81:106-113. https://doi.org/proxy.lib.umich.edu/10.1016/j.ecoenv.2012.04.024.

Frau, D. 2022. Grazing impacts on phytoplankton in South American water ecosystems: a synthesis. Hydrobiologia 849(4):833-860. https://doi.org/10.1007/s10750-021-04748-x.

Frau, D., F.R. Molina, M. Devercelli, and S. José de Paggi. 2012. The effect of an invading filter-feeding bivalve on a phytoplankton assemblage from the Paraná system: a mesocosm experiment. Marine and Freshwater Behaviour and Physiology 45(5):303-316.

Gangi, D., M.S. Plastani, C. Laprida, A. Lami, N. Dubois, F. Bordet, C. Gogorza, D. Frau, and P. de Tezanos Pinto. 2020. Recent cyanobacteria abundance in a large sub-tropical reservoir inferred from analysis of sediment cores. Journal of Paleolimnology 63(3):195-209. https://doi.org/10.1007/s10933-020-00110-8.

Gattas, F., M. Espinosa, P. Babay, H. Pizarro, and D. Cataldo. 2020. Invasive species versus pollutants: Potential of *Limnoperna fortunei* to degrade glyphosate-based commercial formulations. Ecotoxicology and Environmental Safety 201:110794. https://doi.org/10.1016/j.ecoenv.2020.110794.

Gazulha, V., M.C.D. Mansur, L.F. Cybis, and S.M.F.O. Azevedo. 2012. Grazing impacts of the invasive bivalve *Limnoperna fortunei* (Dunker, 1857) on single-celled, colonial and filamentous cyanobacteria. Brazilian Journal of Biology 72(1):33-39. https://doi.org/10.1590/S1519-69842012000100004.

Ghabooli, S., A. Zhan, P. Sardina, E. Paolucci, F. Sylvester, P.V. Perepelizin, E. Briski, M.E. Cristescu, and H.J. MacIsaac. 2013. Genetic diversity in introduced Golden Mussel populations corresponds to vector activity. PLoS ONE 8 (3): e59328.

Gonzalez-Bergonzoni, I., I. Silva, F.T. de Mello, A. D'Anatro, L. Boccardi, S. Stebniki, E. Brugnoli, G. Tesitore, N. Vidal, and D.E. Naya. 2020. Evaluating the role of predatory fish controlling the invasion of the asian golden mussel *Limnoperna fortunei* in a subtropical river. Journal of Applied Ecology 57(4):717-728. https://doi.org/10.1111/1365-2664.13573.

Goto, Y. 2002. Behavior of nuisance mussel, *Limnoperna fortunei,* in water supply facilities. Water Science and Technology 46(11-12):45-50. http://wst.iwaponline.com.proxy.lib.umich.edu/content/46/11-12/45.

Jiahao, Z. 2022. Impact of golden mussel (*Limnoperna fortunei*) colonization on bacterial communities and potential risk to water quality. Ecological Indicators 144:109499. https://doi.org/10.1016/j.ecolind.2022.109499.

Karatayev, A.Y., D. Boltovskoy, D.K. Padilla, and L.E. Burlakova. 2007. The invasive bivalves *Dreissena polymorpha* and *Limnoperna fortunei*: Parallels, contrasts, potential spread and invasion impacts. Journal of Shellfish Research 26(1):205-213. https://doi.org/10.2983/0730-8000(2007)26[205:TIBDPA]2.0.CO;2.

Karatayev, A.Y., L.E. Burlakova, V.A. Karatayev, and D. Boltovskoy. 2010. *Limnoperna fortunei* versus *Dreissena polymorpha*: population densities and benthic community impacts of two invasive freshwater bivalves. Journal of Shellfish Research 29(4):975-984. https://doi.org/10.2983/035.029.0432.

Magara, Y., Matsui, Y., Goto, Y., and Yuasa, A. 2001. Invasion of the non-indigenous nuisance mussel, *Limnoperna fortunei,* into water supply facilities in Japan. Journal of Water Supply: Research and Technology - Aqua 50:113-124. https://d3pcsg2wjq9izr.cloudfront.net/files/5302/articles/9774/Invasionofthenon-indigenous.pdf.

Molloy, D.P., A.Y. Karatayev, L.E. Burlakova, d.P. Kurandina, and F. Laruelle. Natural enemies of Zebra Mussels: predators, parasites, and ecological competitors. Reviews in Fish Biology and Fisheries 5: 27-97 (1997).

Morales, Y., L.J. Weber, A.E. Mynett, and T.J. Newton. 2007. Simulating the effect of invasive species on native freshwater mussel communities. International Journal of River Basin Management 5(4):267-277. https://doi.org/10.1080/15715124.2007.9635326.

Nakano, D., T. Kobayashi, N. Endo, and I. Sakaguchi. 2011. Growth rate and settlement of *Limnoperna fortunei* in a temperature reservoir. Journal of Molluscan Studies 77(2):142-148.

Ogawa, K., T. Nakatsugawa, and M. Yasuzaki. 2004. Heavy metacercarial infections of cyprinid fishes in Uji River. Fisheries Science 70:132-140. https://doi.org/10.1111/j.1444-2906.2003.00781.x.

Oliveira, M.D., D.F. Calheiros, C.M. Jacobi, and S.K. Hamilton. 2011. Abiotic factors controlling the establishment and abundance of the invasive golden mussel *Limnoperna fortunei*. Biological Invasions 13(3):717-729. https://doi.org/10.1007/s10530-010-9862-0.

Paolucci, E.M., D.H. Cataldo, C.M. Fuentes, and D. Boltovskoy. 2007. Larvae of the invasive species *Limnoperna fortunei* (Bivalvia) in the diet of fish larvae in the Parana River, Argentina. Hydrobiologia. 589: 219-233.

Paoulucci, E.M., E.V. Thuesen, D.H. Cataldo, and D. Boltovskoy. 2010. Veligers of an introduced bivalve, *Limnoperna fortunei*, are a new food resource that enhances growth of larval fish in the

Paraná River (South America). Freshwater Biology 55(9):1831-1844. http://onlinelibrary.wiley.com/doi/10.1111/j.1365-2427.2010.02418.x/full.

Paolucci, E.M., V. Leites, D.H. Cataldo, and D. Boltovskoy. 2017. Veligers of the invasive bivalve *Limnoperna fortunei* in the diet of indigenous fish larvae in a eutrophic subtropical reservoir. Austral Ecology 42(7):759-771. https://doi.org/10.1111/aec.12493.

Ricciardi, A. 1998. Global range expansion of the Asian mussel *Limnoperna fortunei* (Mytilidae): another fouling threat to freshwater systems. Biofouling 13(2):97-106. https://doi.org/10.1080/08927019809378374.

Rojas Molina, F., S.J. de Paggi, and D. Frau. 2012. Impacts of the invading golden mussel *Limnoperna fortunei* on zooplankton: a mesocosm experiment. Zoological Studies 51(6):733-744.

Rojas Molina, F., J.C. Paggi, and M. Devercelli. 2010. Zooplanktophagy in the natural diet and selectivity of the invasive mollusk *Limnoperna fortunei*. Biological Invasions 12(6):1647-1659. https://doi.org/10.1007/s10530-009-9578-1.

Rojas Molina, F., and V. Williner. 2013. First record of the non-indigenous mussel *Limnoperna fortunei* (Bivalvia, Mytilidae) as an epibiont of the crab *Trichodactylus borellianus* (Decapoda, Trichodactylidae). Crustaceana 86 (6): 682-692.

Sardiña, P., E. Chaves, and M. Marchese. 2011. Benthic community responses to invasion by the golden mussel, *Limnoperna fortunei* Dunker: biotic homogenization vs environmental driving forces. Journal of the North American Benthological Society 30(4):1009-1023. http://www.journals.uchicago.edu.proxy.lib.umich.edu/doi/10.1899/10-170.1.

Sene, A.M., D.M. Rosa, S.M.M. Gutierre, and P.S. Pompeu. 2021. Freshwater mollusks as proxies for assessing agrochemicals hazards in Volta Grande Reservoir, Brazil. Ambiente e Agua - An Interdisciplinary Journal of Applied Science 16(3):e2681. https://doi.org/10.4136/ambi-agua.2681.

Silva, F.A.e., and A. Giani. 2018. Population dynamic of bloom-forming *Microcystis aeruginosa* in the presence of the invasive bivalve *Limnoperna fortunei*. Harmful Algae 73:148-156. https://doi.org/10.1016/j.hal.2018.02.007.

Silva, I., D. Naya, F.T. de Mello, A. D'Anatro, G. Tesitore, C. Clavijo, and I. Gonzalez-Bergonzoni. 2021. Fish vs. Aliens: predatory fish regulate populations of *Limnoperna fortunei* mitigating impacts on native macroinvertebrate communities. Hydrobiologia 848(9):2281-2301. https://doi.org/10.1007/s10750-020-04421-9.

Spaccesi, F.G., and A.R. Capitulo. 2012. Benthic communities on hard substrates covered by *Limnoperna fortunei* Dunker (Bivalvia, Mytilidae) at an estuarine beach (Río de la Plata, Argentina). Journal of Limnology 71(1):144-153. https://doi.org/10.4081/jlimnol.2012.e15.

SPIC Brasil. 2021. Mexilhão-dourado (Golden mussel). https://www.spicbrasil.com.br/en/destaque/mexilhao-dourado-golden-mussel/. Created on 06/06/2021.

Sylvester, F., J. Dorado, D. Boltovskoy, A. Juarez, and D.H. Cataldo. Filtration rates of the invasive pest bivalve *Limnoperna fortunei* as a function of size and temperature. Hydrobiologia 534: 71-80 (2005).

Sylvester, F., D.H. Cataldo, C. Notaro, and D. Boltovskoy. 2013. Fluctuating salinity improves survival of the invasive freshwater golden mussel at high salinity: implications for the introduction of aquatic species through estuarine ports. Biological Invasions 15(6):1355-1366. https://doi.org/10.1007/s10530-012-0373-z.

Tokumon, R., D. Boltovskoy, and D. Cataldo. 2018. Effects of the invasive freshwater mussel *Limnoperna fortunei* on sediment properties and accumulation rates. Journal of Geophysical Research-Biogeosciences 123(6):2002-2017. https://doi.org/10.1029/2018jg004399.

United States Environmental Protection Agency (EPA). 2015. Analysis of Ballast Water Discharges into the Great Lakes from Overseas Vessels from 2010 to 2013. Office of Water / Office of Wastewater Management / Water Permits Division, Washington, DC. https://www.epa.gov/vessels-marinas-and-ports/ballast-water-discharges-great-lakes-overseas-vessels.

Vargas, R.P.F., J.F. Saad, M. Graziano, M.D. Afonso, I. Izaguirre, and D. Cataldo. 2019. Bacterial composition of the biofilm on valves of *Limnoperna fortunei* and its role in glyphosate degradation in water. Aquatic Microbial Ecology 83(1):83-94. https://doi.org/10.3354/ame01907.

Weigand, C. 2019. Strategies and consequences of indigenous and invasive freshwater mussels living in cyanobacterial and anthropogenic impacted waters. Pages 93-128 in Gross, E. and Garric, J, eds. Ecotoxicology: New challenges and new approaches. Iste Ltd. London.

Xia, Z.Q., J.R. Barker, A.B. Zhan, G.D. Haffner, and H.J. MacIsaac. 2021. Golden mussel (*Limnoperna fortunei*) survival during winter at the northern invasion front implies a potential high-latitude distribution. Diversity and Distributions 27(8):1422-1434. https://doi.org/10.1111/ddi.13289.

Yao, G.Y., M.Z. Xu, and X.H. An. 2017. Concrete deterioration caused by freshwater mussel *Limnoperna fortunei* fouling. International Biodeterioration & Biodegradation 121:55-65. https://doi.org/10.1016/j.ibiod.2017.03.011.

Zhang, H., E.S. Rutherford, D.M. Mason, M.E. Wittmann, D.M. Lodge, X. Zhu, T.B. Johnson, and A. Tucker. 2019. Modeling potential impacts of three benthic invasive species on the Lake Erie food web. Biological Invasions 21:1697-1719. https://doi.org/10.1007/s10530-019-01929-7.

Zhang, R., B. Cui, Z. Sun, W. Zhou, P. Ren, Z. Du, N. He, and J. Lu. 2017. Feeding ecology of *Limnoperna fortunei* in southern China: Insights from stable isotopes and fatty-acid biomarkers. Journal of Freshwater Ecology 32(1):761-772. https://doi.org/10.1080/02705060.2017.1399930.